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Code: ITA0502
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Activity 2.
Scenario - Based

① CCTV image with salt and pepper noise?
Technique : Median filter
Reason : Median filtering is a non-linear spatial filter specifically effective at removing impulsive (salt and pepper) noise without blurring sharp edges.

② Satellite image corrupted by Gaussian noise
Technique : Gaussian Filter
Reason : Gaussian noise is additive and normally distributed.

③ Medical x-ray image.
Technique : Bilateral filter
Reason : Bilateral filtering considers both geometric distance and pixel intensity differences effectively smoothing random-noise.

④ Scanned document with scatted black and white dots
Technique: Morphological operation.
Reason: Morphological opening removes small white forground spots

⑤ Low-Light image heavy grain and no-clear boundaries.
Technique: Non-Local means (NLM) Densing
Reason: Averages similar patterns across the entire

image to clear heavy grain

6) Foggy road image with very Low constrast
Technique: Dark Channel Prior
Reason: Removes atmospheric haze and boosts cont

7) underwater image that appears dull and washed o
Technique: CLAHE
Reason: Enhances local contrast without over-amplif
noise.

8) Chest x-ray with poor visibility of internal
Structures.
Technique: constrast strecteching / CLAHE
Reason: Expands dynamic range to highligh
subtle tissue and bone details

9) Night-time traffic image Locking intensity
Variation.
Technique: Gamma correction.
Reason: Brightness dark shadows while keepi
headlights from blowing out.

10) A grayscale image uses only a narrow intensi
range.
Technique: Contrast strectching
Reason: spreads narrow intensity values across
the full range (0-255) for better clarity